

Worksheet 3 Answers

Deutsch–Jozsa Algorithm

December 3, 2024

1 Problem 1

Recall for each problem that $f(x)$ is *binary* - it only takes on 0 or 1. If the idea of $\oplus$ being mod two doesn't make sense, you can also think of it as the aforementioned CNOT on $f(x)$ - if x is 0, keep the result of $f(x)$; otherwise, if $x = 1$, then flip the result of $f(x)$.

$$(a) \quad U_f |x\rangle |0\rangle = |x\rangle |f(x)\rangle$$

$$(b) \quad U_f |x\rangle |1\rangle = |x\rangle |\overline{f(x)}\rangle$$

2 Problem 2

(a) Use the substitution $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ for $|-\rangle$, and plug in values from problem 1:

$$U_f |x\rangle |-\rangle = U_f |x\rangle \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}}(U_f |x\rangle |0\rangle - U_f |x\rangle |1\rangle) \quad (1)$$

$$= \frac{1}{\sqrt{2}}(|x\rangle |f(x)\rangle - |x\rangle |\overline{f(x)}\rangle) = \frac{1}{\sqrt{2}} |x\rangle (|f(x)\rangle - |\overline{f(x)}\rangle) \quad (2)$$

(b) If $f(x) = 0$, we substitute 1 for $\overline{f(x)}$:

$$\frac{1}{\sqrt{2}} |x\rangle (|0\rangle - |1\rangle) = |x\rangle \left(\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \right) = |x\rangle |-\rangle$$

If $f(x) = 1$, we substitute 0 for $\overline{f(x)}$:

$$\frac{1}{\sqrt{2}} |x\rangle (|1\rangle - |0\rangle) = |x\rangle \left(\frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) \right) = -|x\rangle \left(\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \right) = -|x\rangle |-\rangle$$

(c) So, we want to keep $|-\rangle$ if $f(x) = 0$ and flip the sign to $-|-\rangle$ if $f(x) = 1$. As the hint states, we want to use $(-1)^{f(x)}$ to accomplish this (this is 1 if $f(x) = 0$, and -1 if $f(x) = 1$):

$$U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle$$

Recall that $f(x)$ is binary in this scenario (it is 0 or 1).

3 Problem 3

- (a) From the diagram, $|\psi_1\rangle$ is the result of calling the hadamard gate on both a $|0\rangle$ and a $|1\rangle$. The result of a Hadamard on $|0\rangle$ is $|+\rangle$ and the result of a Hadamard on $|1\rangle$ is $|-\rangle$. The mathematical result of applying the Hadamard to both $|0\rangle$ and $|1\rangle$, then, is the following:

$$|\psi_1\rangle = H|0\rangle \otimes H|1\rangle = |+\rangle |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) |-\rangle \quad (3)$$

$$(4)$$

Here is the proof of $H|0\rangle = |+\rangle$ and $H|1\rangle = |-\rangle$:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \implies H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (5)$$

$$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle \quad (6)$$

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} \quad (7)$$

$$= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle \quad (8)$$

- (b) Now, according to the diagram, we apply U_f with $|+\rangle$ as the input and $|-\rangle$ as the control. As per our results from problem 2(b):

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (9)$$

$$U_f |+\rangle |-\rangle = U_f \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) |-\rangle \right) = \frac{1}{\sqrt{2}} (U_f |0\rangle |-\rangle + U_f |1\rangle |-\rangle) \quad (10)$$

$$= \frac{1}{\sqrt{2}} \left((-1)^{f(0)} |0\rangle |-\rangle + (-1)^{f(1)} |1\rangle |-\rangle \right) = \frac{1}{\sqrt{2}} ((-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle) |-\rangle \quad (11)$$

This is an ugly final answer, but...

- (c) Analyze the first qubit with all constants extrapolated out (namely, $\frac{1}{\sqrt{2}}$ or any negatives).

$$((-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle) \quad (12)$$

If $f(0) = f(1)$ (constant function), then you'll get a $|+\rangle$ state. If $f(0) \neq f(1)$ (balanced function), then you'll get a $|-\rangle$ state.

- (d) Applying the Hadamard gate onto $|+\rangle$, we get $H|+\rangle = |0\rangle$. Correspondingly, $H|-\rangle = |1\rangle$. This is just a nicer format for our final answer. $|0\rangle$ corresponds with the function being constant, and $|1\rangle$ corresponds with the function being balanced.